Who Buys the Exit? Domestic Prosecution Risk and the Purchase of Foreign Citizenship

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Abstract

Who buys their way out of an authoritarian state's enforcement? Using a newly compiled register of all 4,167 foreign naturalizations granted by Cambodia (2000–2023), I study cross-border investment naturalization as an observable margin of elite exit. In 2018, a Cambodian reform that fast-tracked naturalization while raising its price sharply expanded foreign naturalization. I exploit this common host-country opening to ask which Chinese provinces took it up. Using the cross-province intensity of Xi Jinping's anti-corruption campaign as a measure of prosecution risk, I find that take-up rose disproportionately from high-enforcement provinces: a one–standard-deviation increase in pre-2018 enforcement raises post-2018 naturalizations by a factor of roughly 2.6. The estimate survives a score wild-cluster bootstrap and controls for provincial wealth, diaspora networks, province corruption history, and telecom-fraud origin—addressing whether high enforcement provinces merely supply Cambodia's rising scam economy—and reproduces at the individual level (2.30 times higher odds of post-opening entry, driven by enforcement, not wealth). That take-up surged despite a higher price points to demand for a credible exit rather than a discount—rare individual- and regional-level evidence consistent with purchased citizenship as a hedge against domestic prosecution and expropriation risk.

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1 Introduction

A growing share of the world's wealthy hold more than one passport, and a growing number of states sell citizenship outright. The "golden passport" industry (citizenship- and residence-by-investment (CBI/RBI) programs) now spans dozens of countries and intermediates billions of dollars in cross-border payments (Surak 2023). Yet the empirical literature on these programs is constrained by a fundamental data limitation: it observes *programs* and aggregate flows, never the *individuals* who buy in. As a result, the central behavioral question—*who* purchases a second citizenship, and *why*—has been addressed only through theory and anecdote to the best of the author's knowledge.

This paper opens that black box for one country. I use a newly compiled, individual-level register of every foreigner naturalized by the Kingdom of Cambodia between 2000 and 2023 collected from official government documentation by the Kouprey (2026). The data record each person's nationality, place of birth, sex, date of birth, exact date of naturalization, and adopted name. Cambodia is an instructive case: its 2018 reform to the Law on Nationality was followed by a sharp rise in investment naturalization. Take-up since the scheme began in 1996 (but publicly recorded starting 2000) has been overwhelmingly Chinese—2,321 of 4,167 naturalizations—and extraordinarily concentrated in a handful of Chinese provinces, with Fujian alone accounting for 39% of all Chinese naturalizers.

This study asks whether this purchased citizenship functions as a hedge against domestic governance risk. The identification challenge is that Cambodia's 2018 opening is a single host-country shock common to every origin, perfectly absorbed by year fixed effects; it cannot explain *which* origins responded. I therefore turn the question around. Conditional on the exit route being open, do individuals from Chinese provinces under heavier domestic prosecution pressure take it up at higher rates? I measure prosecution pressure using the differential provincial intensity of Xi Jinping's 2012–2016 anti-corruption campaign, from the Wang and Dickson (2020) register of nearly 19,000 investigated officials. The design is a continuous-treatment difference-in-differences: a province's pre-determined enforcement "dose" interacted with the post-2018 opening, with province and year fixed effects.

The central finding is that a one–standard-deviation higher pre-2018 enforcement dose raises a province's post-2018 naturalizations by a factor of about 2.6 (Poisson pseudo-maximum-likelihood; (Santos Silva and Tenreyro 2006)). The estimate is robust to a score wild-cluster bootstrap appropriate for our 28 provincial clusters (Cameron et al. 2008; MacKinnon 2023), and survives a horse race against provincial wealth and historic emigration networks. The divergence appears overwhelmingly *after* the 2018 opening: a placebo interaction with the 2013–2016 campaign years is statistically insignificant and is in any case less than a quarter the size of the post-2018 effect. Critically, the enforcement dose is distinct from wealth (cross-province correlation −0.39) and only moderately correlated with diaspora networks (0.50); high-dose provinces include poor interior regions (Shanxi, Guizhou, Ningxia), not only rich coastal ones.

The analysis then moves to the individual level, which directly addresses the ecological-inference concern inherent in any province-level design. Among Chinese naturalizers, those from high-enforcement provinces are 2.31 times more likely to belong to the post-2018 cohort, controlling for age, sex, name adoption, and origin-province income—and origin income enters *negatively*, confirming that the margin is enforcement rather than affluence. The post-opening cohort is also younger and more male, consistent with a business/asset-holding exit profile rather than family or retirement migration.

The results can be interpreted as a newly observable *window* onto elite capital flight, not a measure of its total volume. Cambodian naturalization is a thin, selected slice of Chinese elite exit; its value is that it is observable at the level of the individual and the home province, which standard capital-flight proxies (balance-of-payments errors and omissions, outward FDI) are not. The mechanism is given human form by documented cases—fugitives, money launderers, and sanctioned operators who acquired Cambodian citizenship shortly before prosecution at home (Kouprey 2026).

2 Related Literature

This paper sits at the intersection of five strands of literature. First, the paper directly touches the small but growing body of work studies citizenship- and residence-by-investment programs as markets in membership. Surak (2023) provides the definitive account of the industry and its clientele, and Surak (2021) analyzes the sale of citizenship as a form of millionaire mobility; policy and legal work documents the proliferation of programs and their money-laundering and security risks (Shachar and Bauböck 2014). International organizations have begun to quantify program adoption and aggregate effects (International Monetary Fund 2025). What this literature lacks, uniformly, is individual-level data on the buyers: it observes programs and aggregate flows but never the people who naturalize. This paper's contribution is to open that black box for one country, linking individual purchases to the buyers' regions of origin.

Second, the paper adds to the capital flight, offshore wealth, and regulatory evasion literature. A large literature measures the movement of *money* out of countries with weak property-rights protection. Le and Zak (2006) link political risk to capital flight; Gunter (2004) documents large and rising capital flight from China; and a more recent strand quantifies the offshore wealth held by households in tax havens (Zucman 2013; Alstadsæter et al. 2018). A parallel political-economy literature studies how kleptocratic elites use the global financial and professional system to move and shelter assets (Sharman 2017; Cooley and Sharman 2017). This paper adds the *personal* margin that complements hidden money: the acquisition of a foreign citizenship, observed at the level of the individual and the home province, and tied to a specific domestic governance shock.

Third, this research speaks to elite mobility as a response to policy. A public-economics literature shows that the geographic location of the wealthy responds to policy, principally taxation: top earners and star scientists move in response to tax differentials (Moretti and Wilson 2017; Kleven et al. 2020), as do high-skilled inventors (Akcigit et al. 2016). This work is set almost entirely in democracies and concerns taxation. I extend the logic to an authoritarian setting and to a different push factor—the risk of prosecution and expropriation—and I study an extreme form of exit, the purchase of an outside option on citizenship itself.

Fourth, this paper looks at anti-corruption policy in China. Xi Jinping's campaign has been studied intensively, including through the investigation microdata used in this analysis (Wang and Dickson 2020). Research documents its domestic consequences for corporate investment and valuations, luxury consumption, and official behavior (Griffin et al. 2022;

Lin et al. 2016), and debates whether it is a genuine clean-up or a tool of factional consolidation. This literature is almost entirely *inward-looking*—it studies effects within China. I document an outward, general-equilibrium margin: enforcement at home is associated with the displacement of wealthy individuals abroad through a market in foreign citizenship.

Finally, the geographic findings connect to the economics of migrant selection and networks. Classic models predict selection on the returns to mobility (Borjas 1987), and a large literature shows that pre-existing diaspora networks shape who migrates and to where (Munshi 2003; Beine et al. 2011), including the ethnic-Chinese networks that structure trade and movement across Southeast Asia (Rauch and Trindade 2002); the Fujianese case is especially well documented (Pieke et al. 2004). I use historic emigration networks (*qiaoxiang*) both as a control and as an explanation for the extreme concentration of take-up, and I show that the enforcement channel operates over and above them.

3 Institutional Background and Data

3.1 Cambodian investment naturalization

Cambodia's Law on Nationality, first enacted in 1996, permits naturalization through investment or donation. The investment pathway was substantially reformed in 2018: the Law on Nationality was adopted by the National Assembly on 31 May 2018, approved by the Senate on 11 June 2018, and promulgated by Royal Kram NS/RKM/0618/008 on 21 June 2018. The reform made three changes that matter here. First, it *fast-tracked* the route, cutting the legal-residence requirement from seven years for ordinary naturalization to *twelve months* for naturalization by investment and *six months* for naturalization by cash donation (Arts. 21–22). Second, it formalized investment naturalization as an explicitly *discretionary* pathway—"not a right of the applicant" but a concession of the state, with conditions waivable by decision of the Prime Minister (Art. 18). Third, it delegated the monetary threshold to sub-decree; the operative amounts and procedures were set by Sub-decree No. 112 ANKr.BK of 30 August 2018, which *raised* the headline investment requirement—from roughly 1.25 billion riel (about US$312,000) to nearly US$1 million (Phnom Penh Post 2018).

Two features of this history matter for what follows. First, the route became operationally usable only once the implementing sub-decree took effect at the end of August 2018, which aligns with the timing of take-up in our data: counts are flat through mid-2018 and accelerate only from late 2018, peaking in 2019. Second, the 2018 reform made citizenship *faster but more expensive*—take-up surged *despite* a higher headline price, a pattern more consistent with demand for a credible exit option than with bargain-seeking. The publication of naturalization data in the Royal Gazette ceased after 2023 amid money-laundering scandals (Organized Crime and Corruption Reporting Project 2024); our panel therefore ends in 2023, and I treat the 2024 collapse to near-zero as administrative right-censoring rather than a behavioral decline. For a more comprehensive summary of the law and its legal history, see Appendix A.

3.2 The naturalization microdata

Kouprey (2026) compiled all 4,167 naturalizations recorded in the Royal Gazette, 2000–2023, through three months of manual collation, translation, and double-checking against the original gazettes. Each record contains nationality, place of birth, sex, date of birth, date of naturalization, and any adopted Cambodian name. Chinese nationals account for 2,321 records (56%), followed by Taiwanese (369) and Korean (303). The data are male-dominated (86%), consistent with an investor rather than a family-migration profile.

For the analysis I restrict to Chinese nationals whose birthplace identifies a mainland provincial-level division. Of 2,321 Chinese records, 94.7% map cleanly to one of the 31 mainland divisions; the remainder are Hong Kong/Macau SARs (excluded), foreign birthplaces, or ambiguous strings (Appendix B details the coding). Origin is extraordinarily concentrated: Fujian supplies 39% of mainland Chinese naturalizers, the top three provinces 58%, with a Gini coefficient of 0.73 across provinces.

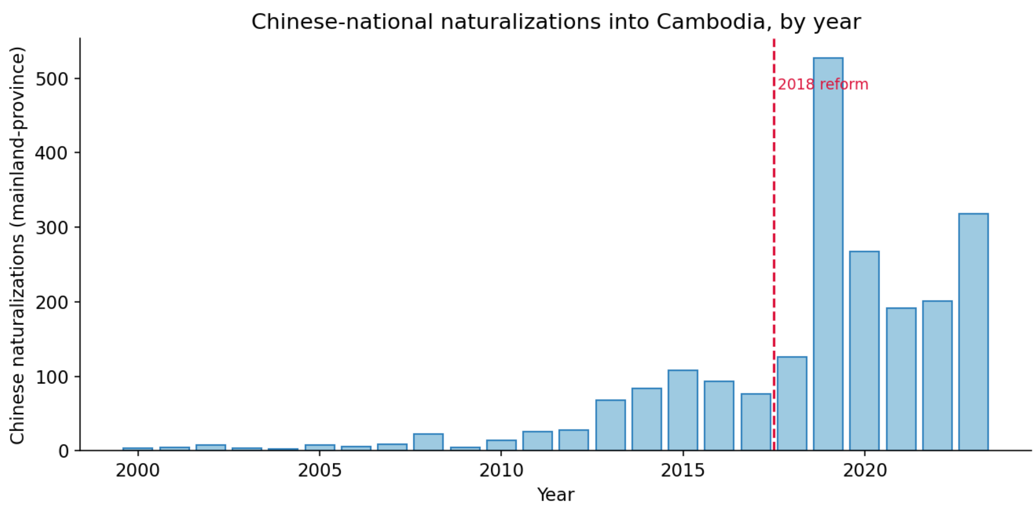

Figure 1: Chinese-national naturalizations into Cambodia by year, 2000–2023 (mainland provinces, by date of naturalization). Take-up is negligible before 2013, rises gradually through the mid-2010s, and jumps sharply after the 2018 reform (dashed line), peaking in 2019.

Figure 1 shows the time series of Chinese-national naturalizations into Cambodia from 2000 to 2023. Even before the 2018 reform, there was a gradual rise in naturalizations, only breaking 100 in 2015 (108) and even dissipating after, falling back to 93 in 2016 and 76 in 2017. Following the reform, there is a sharp spike in naturalizations, from 126 in 2018 to 527 in 2019—a roughly fourfold jump in a single year, and nearly five times the pre-reform peak. The series then recedes through the COVID-19 years, when Cambodia's borders were largely closed (267 in 2020, and around 190–200 in 2021–2022), before rebounding to 318 in 2023. The contrast is stark: the 569 naturalizations recorded across the eighteen pre-reform years (2000–2017) are dwarfed by the 1,630 recorded in the six years after. This aggregate surge is common to the host-country opening—precisely what the year fixed effects absorb—and is not itself the object of estimation; the analysis that follows instead exploits variation *across Chinese provinces* in who took up citizenship against this common shock.

3.3 The outcome panel

I aggregate the microdata to a balanced panel of 31 mainland provinces $\times$ 24 years (2000–2023), with naturalization counts and structural zeros where no naturalizations occur. Of the 31 mainland provinces, three — Ningxia, Xinjiang, and Tibet — record zero naturalizations in every year and are perfectly separated by their province fixed effect; PPML drops them, leaving 28 provinces (672 province-years) and 28 clusters for inference. Figure 2 plots the trajectories: a steep, Fujian-led surge after the 2018 opening that dwarfs all other provinces. Figure 3 shows the same concentration across the province $\times$ year grid.

3.4 The treatment: provincial enforcement intensity

The paper measures province-level prosecution risk using the Wang and Dickson (2020) China Corruption Investigations Dataset, which records 18,947 officials investigated during the anti-corruption campaign, dated and geolocated by province (the campaign concentrated in 2013–2016). We construct each province's enforcement *dose* as investigations per million residents (2012–2016 cumulative; population from the 2010 Population Census, (National Bureau of Statistics of China 2012)), standardized. The highest-intensity provinces are a mix of coastal and interior regions—Hainan, Shanxi, Guizhou, Guangxi, Yunnan, Fujian—and the dose is only *moderately* correlated with diaspora networks (0.50) and *negatively* correlated with provincial GDP (-0.39); I condition on both interacted with Post-2018 throughout.

3.5 Controls and data sources

Provincial nominal GDP, 2000–2023, is from the National Bureau of Statistics of China (China Statistical Yearbook), with the 2021–2023 figures incorporating the 2023 Fifth National Economic Census revision (National Bureau of Statistics of China 2024). Provincial population (the dose denominator) is from the 2010 Population Census of China (National Bureau of Statistics of China 2012). The diaspora-network control, *qiaoxiang*, is an indicator for the historically documented overseas-Chinese-sending provinces—Guangdong, Fujian, Hainan, Guangxi, and Zhejiang—the standard emigrant-origin regions in the migration literature (Pieke et al. 2004; Rauch and Trindade 2002). I use a classification rather than counts because province-by-province *counts* of overseas Chinese are not available from any single official source, whereas the qualitative qiaoxiang classification is well established. Table 1 lists every variable and its source, and Table 2 reports summary statistics.

Table 1: Variables and data sources

Variable	Description	Source
Naturalizations	Cambodian naturalizations, person-level	Kouprey (2026)
Enforcement dose	Officials investigated 2012–16, per capita	Wang and Dickson (2020)
Population	Provincial population (dose denominator)	NBS2012
Qiaoxiang	Documented overseas-Chinese-sending provinces	Pieke et al. (2004)
GDP	Provincial nominal GDP, 2000–2023	NBS2024
Corruption index	MIMIC provincial corruption	H. Chen et al. (2018)

Notes. NBS2012 and NBS2024 stands for National Bureau of Statistics of China (2012) and National Bureau of Statistics of China (2024) respectively.

Table 2: Summary statistics, province-year panel (2000–2023)

Variable	Mean	SD	Min	Max
Naturalizations (count)	2.96	13.69	0.00	263.00
Log provincial GDP	9.33	1.15	5.57	11.83
GDP growth (YoY)	0.12	0.06	−0.04	0.31
Qiaoxiang province (indicator)	0.18	0.38	0.00	1.00
Province-year observations: 672 (28 provinces with positive counts)				

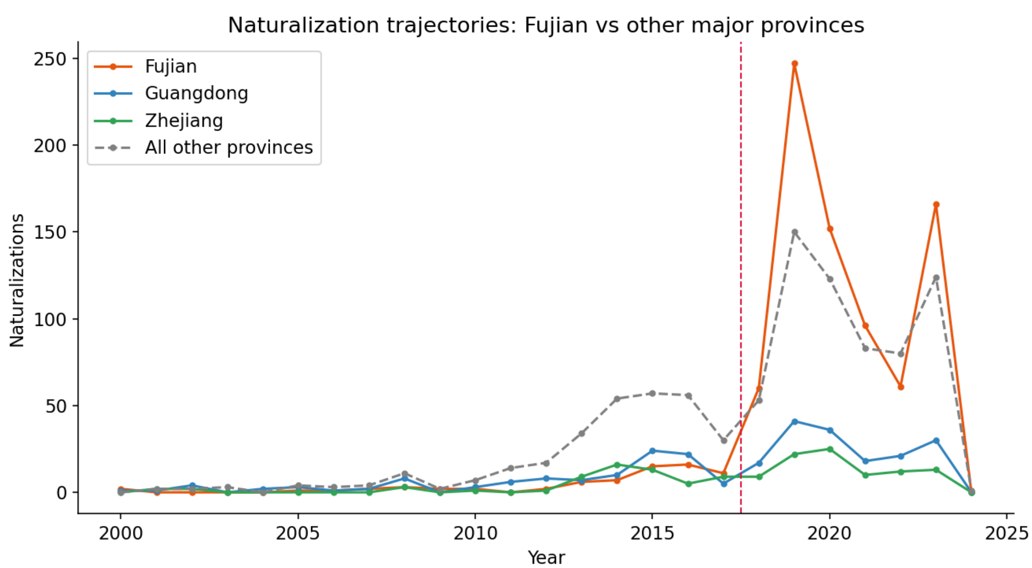

Figure 2: Naturalization trajectories by origin province. Fujian's post-2018 surge (peak 263 in 2019) dwarfs other major provinces and all other provinces combined. Dashed line: 2018 Cambodian opening.

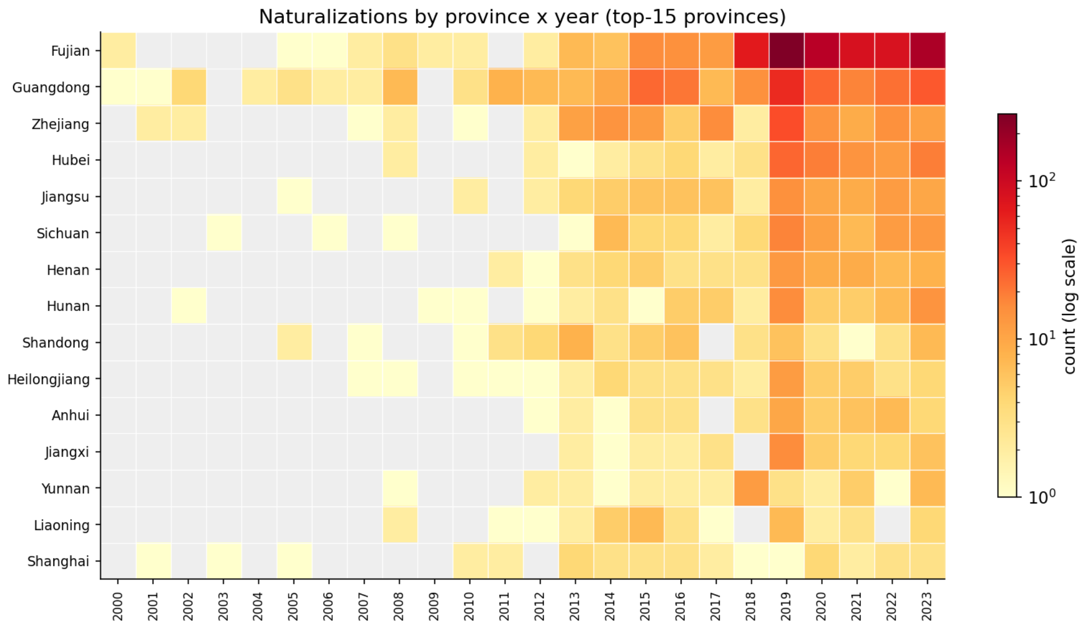

Figure 3: Naturalizations by province × year (top-15 provinces). The intensity is concentrated in Fujian and in the years 2019–2023.

4 Empirical Strategy

4.1 Setup and the identification problem

Cambodia's 2018 reform opened a common exit route for all origins on the same date. In a province panel, such a common shock is absorbed entirely by year fixed effects: it cannot, by construction, explain *which* Chinese provinces responded, and it is in any case plausibly endogenous to Chinese demand. I therefore do not attempt to estimate the effect of the reform. Instead I identify off *differential exposure* to it. Provinces differ in a predetermined characteristic—the intensity of anti-corruption enforcement they experienced during the 2012–2016 campaign—and I ask whether more heavily enforced provinces took up the newly available exit route at higher rates.

4.2 Main specification

Our estimating equation is a continuous-treatment difference-in-differences, estimated by Poisson pseudo-maximum-likelihood (PPML)—the appropriate estimator for a skewed, non-negative count outcome with many zeros (Santos Silva and Tenreyro 2006):

$$\mathbb{E}[\mathrm{Nat}_{pt} \mid \cdot] = \exp\Big(\beta \left(\mathrm{Dose}_p \times \mathrm{Post2018}_t\right) + \mathbf{x}'_{pt}\boldsymbol{\gamma} + \alpha_p + \delta_t\Big), \tag{1}$$

estimated over 28 mainland provinces p (those with positive naturalizations) and years $t = 2000, \ldots, 2023$. The components are:

- Nat_{pt} — the number of Chinese nationals from province p naturalized by Cambodia in year t (the dependent variable);

- Dose_p — the standardized, pre-determined enforcement intensity of province p (anti-corruption investigations per million residents, 2012–2016 cumulative, from (Wang and Dickson 2020));

- $\text{Post2018}_t = \mathbf{1}[t \geq 2018]$ — an indicator for the post-reform period;

- β — the coefficient of interest: the proportional effect on naturalizations of a one–standard-deviation higher enforcement dose, after the exit route opens;

- $\mathbf{x}_{pt}$ — time-varying controls, namely $\text{LogGDP}_{pt} \times \text{Post2018}_t$ (a wealth/capital channel) and $\text{Qiaoxiang}_p \times \text{Post2018}_t$ (a diaspora-network channel);

- $\alpha_p,\ \delta_t$ — province and year fixed effects.

The province fixed effects α_p absorb the *level* of every time-invariant provincial characteristic, including the dose and the qiaoxiang measure themselves; the year effects δ_t absorb the 2018 reform and all nationwide shocks (capital controls, the campaign, the business cycle). Identification of β thus comes purely from the interaction: whether high-dose provinces diverge from low-dose provinces *after* 2018 relative to before. Standard errors are clustered by province.

4.3 Identification and inference

Identification requires that, absent the 2018 opening, high- and low-enforcement provinces would have followed parallel naturalization paths. Three features support this. First, the temporal ordering rules out reverse causality: the dose is fixed in 2012–2016, before the 2018 outcome, so naturalization cannot drive the investigations. Second, the enforcement dose is not a proxy for wealth or diaspora networks—it is negatively correlated with provincial GDP (-0.39) and only moderately with the qiaoxiang indicator (0.50)—and I condition on both channels interacted with Post-2018. Third, I test for pre-existing divergence with a parsimonious placebo: a $\text{Dose}_p \times \mathbf{1}[2013 \leq t \leq 2017]$ interaction added to Eq. (1). As reported in Section 5, this placebo is statistically insignificant at conventional levels (its 95% confidence interval includes zero), while the post-2018 interaction is more than four times larger and strongly significant; I cannot reject parallel pre-trends, though the positive placebo point estimate counsels some caution. The remaining threat is that high-enforcement provinces differ on some omitted dimension that independently raised their post-2018 emigration; the horse race and the placebo mitigate but cannot eliminate this, so the estimates are framed as *consistent with* a causal interpretation rather than as a clean experiment.

With only 28 provincial clusters, asymptotic cluster-robust standard errors over-reject. The paper therefore reports score wild-cluster bootstrap p-values with Rademacher weights (Cameron et al. 2008; MacKinnon 2023), the recommended procedure for few clusters and nonlinear models, for both the aggregate and individual-level results.

5 Results

5.1 Selection: who naturalizes

Table 3 documents how the composition of Chinese naturalizers shifted with the 2018 opening. The post-2018 cohort is 7.4 percentage points more male and 5.5 years younger than the pre-2018 cohort, and slightly less likely to adopt a distinct Cambodian name. This is the demographic signature of business/asset-holding exit rather than family or retirement migration, and it is observable only because the data are individual-level.

Table 3: Selection among Chinese naturalizers (OLS, robust SE)

	Male (1)	Age at nat. (2)	Adopted name (3)
Post-2018 cohort	0.074***	−5.498***	−0.041
	(0.018)	(0.477)	(0.022)
High-qiaoxiang origin	0.080***	−2.399***	−0.181***
	(0.014)	(0.397)	(0.019)
Observations	2,313	2,315	2,321

Notes. OLS with heteroskedasticity-robust SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

5.2 The main result: enforcement and exit

Table 4 reports the DiD. In column (1), a one–standard-deviation higher enforcement dose raises post-2018 naturalizations by $\exp(1.032) \approx 2.8$. Adding wealth (column 2) and diaspora networks (column 3) interacted with Post-2018 leaves the enforcement effect strong: $\hat{\beta} = 0.941$ ($p < 0.001$), i.e. a factor of $\exp(0.941) \approx 2.6$ per standard deviation. Notably, LogGDP × Post2018 enters *negatively*: conditional on enforcement and networks, richer provinces send fewer, not more. The qiaoxiang network channel is positive but, with the documented qiaoxiang indicator, statistically insignificant (0.14, $p \approx 0.28$); it does not displace enforcement.

Table 4: Enforcement dose × Post-2018: continuous-treatment DiD (PPML)

Outcome: naturalizations	(1)	(2)	(3)
Enforcement dose × Post-2018	1.032**	1.097**	0.941***
	(0.372)	(0.371)	(0.254)
Log GDP × Post-2018		−0.471	−0.602*
		(0.247)	(0.279)
Qiaoxiang × Post-2018			0.143
			(0.133)
Province FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	672	672	672

Notes. PPML estimates; cluster-robust SE (province) in parentheses. The preferred specification (3) yields a score wild-cluster bootstrap p-value of < 0.001 for the enforcement interaction (Table 7). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (i.e. a star denotes a 95% confidence interval excluding zero, or better).

5.3 Pre-trends and dynamics

A fully flexible event study interacting the dose with all 24 years is hopelessly over-parameterized for 28 clusters (and the near-empty pre-2008 years generate uninformative noise), so I test pre-trends and dynamics with a parsimonious binned specification (Table 5). Column (1) adds a placebo Dose × (2013–17) interaction to the main DiD, with 2000–2012 as the omitted base. The placebo is statistically insignificant at conventional levels (0.227; 95% CI $[-0.07, 0.52]$), whereas the post-2018 interaction is more than four times larger and strongly significant (1.111, $p < 0.001$): I cannot reject parallel pre-trends, and any divergence during the campaign years is far too small to account for the post-opening jump. Column (2) splits the post period and shows the effect is large and stable across 2018–2020 (1.146) and 2021–2023 (1.067)—a persistent shift, not a transitory spike. High-enforcement provinces diverge sharply only once the exit route opens.

Table 5: Pre-trends and dynamics (PPML, province & year FE; base period 2000–2012)

Outcome: naturalizations	(1) Pre-trend test	(2) Binned dynamics
Dose × (2013–17), placebo	0.227	0.227
	(0.150)	(0.150)
Dose × Post-2018	1.111***	
	(0.282)	
Dose × (2018–20)		1.146***
		(0.294)
Dose × (2021–23)		1.067***
		(0.275)
GDP × Post, Qiaoxiang × Post	Yes	Yes
Province FE, Year FE	Yes	Yes
Observations	672	672

Notes. PPML; cluster-robust SE (province) in parentheses. The placebo campaign-years interaction is not significant at the 5% level (95% CI $[-0.07, 0.52]$) and is less than a quarter the size of the post-2018 effect. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (i.e. a star denotes a 95% confidence interval excluding zero, or better).

5.4 Event-study specification

To trace the response year by year and to test formally for differential pre-trends, I estimate a dynamic version of the main specification. Replacing the single $\text{Dose}_p \times \text{Post2018}_t$ interaction in Eq. (1) with a full set of year-specific dose interactions, the PPML estimating equation is

$$\mathbb{E}[\text{Nat}_{pt} \mid \cdot] = \exp\Big(\sum_{k \neq 2017} \beta_k \left(\text{Dose}_p \times \mathbf{1}[t = k]\right) + \mathbf{x}'_{pt}\boldsymbol{\gamma} + \alpha_p + \delta_t \Big), \tag{2}$$

where $\mathbf{1}[t = k]$ indicates year k and the last pre-reform year, $k = 2017$, is omitted as the reference; each β_k therefore measures the cross-province dose gradient in year k *relative to 2017*. Because naturalizations are negligible before 2013, the early years are pooled into a single interaction $\text{Dose}_p \times \mathbf{1}[t \leq 2012]$ to avoid over-parameterization and the noise of near-empty cells. As in Eq. (1), α_p and δ_t are province and year fixed effects, $\mathbf{x}_{pt}$ collects the wealth (LogGDP_{pt}) and diaspora-network (Qiaoxiang_p) controls interacted with Post-2018, standard errors are clustered by province, and inference uses the score wild-cluster bootstrap.

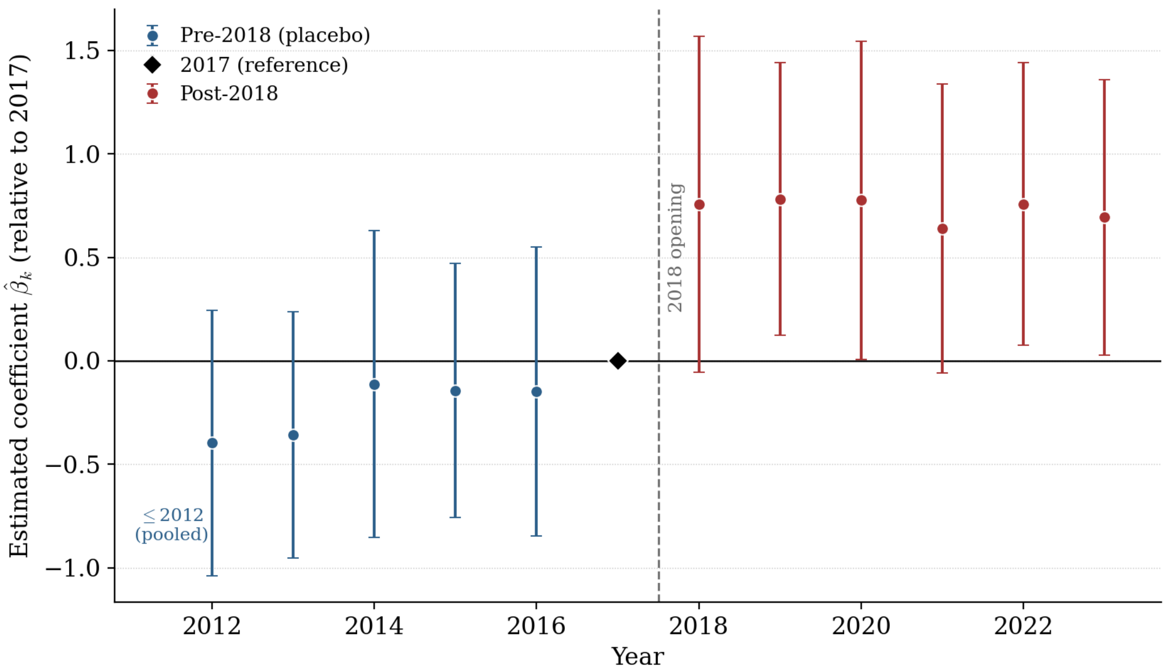

Figure 4: Event study: enforcement dose × year, relative to 2017 (t−1). Points are PPML coefficients; bars are 90% cluster-robust confidence intervals; the pre-2013 years are pooled. Pre-2018 coefficients are jointly zero ($F = 0.81$, $p = 0.56$): no differential pre-trend. The effect emerges at the 2018 opening; 2019, 2020, 2022, and 2023 are individually significant by wild-cluster bootstrap ($p < 0.10$), while 2018 (partial-year) and 2021 (COVID border closures) are imprecise. Significance of the average post-2018 effect rests on the pooled estimate (Table 7, 0.941, $p < 0.001$).

The coefficients in Figure 4 have a transparent reading. The *pre-reform* coefficients ($k \leq 2016$) are placebo estimates: under parallel trends, high- and low-enforcement provinces should not diverge before the route opens, so these should be jointly zero. The *post-reform* coefficients ($k \geq 2018$) trace the dynamic response. The identifying assumption is assessed by a joint Wald test of the null that all pre-2018 interactions vanish,

$$H_0: \ \beta_{\leq 2012} = \beta_{2013} = \beta_{2014} = \beta_{2015} = \beta_{2016} = 0; \tag{3}$$

failure to reject is consistent with no differential pre-trend.

Figure 4 plots the estimated $\{\beta_k\}$ with 90% confidence intervals. The pre-reform coefficients are individually small and jointly indistinguishable from zero (test (3): $F = 0.81$, $p = 0.56$), so the parallel-trends assumption is not rejected. The dose gradient emerges discretely at the 2018 opening and persists through 2023; the post-2018 coefficients for 2019, 2020, 2022, and 2023 are individually significant by the wild-cluster bootstrap ($p < 0.10$), while 2018 (a partial treatment year—the implementing sub-decree took effect only at the end of August) and 2021 (COVID-19 border closures) are less precisely estimated. Significance of the *average* post-2018 effect rests on the pooled estimate of Eq. (1).

5.5 Individual-level evidence

The province-level result is open to an ecological-inference critique: it shows that high-enforcement *provinces* send more naturalizers, not that this gradient also holds across *indi-*

viduals. I provide individual-level corroboration by returning to the microdata. The unit is now an individual Chinese naturalizer i with origin province $p(i)$, and I estimate a logit for the probability that the person belongs to the post-opening cohort:

$$\Pr\Big(\text{Post2018}_i = 1 \,\Big|\, \cdot\,\Big) = \Lambda\Big(\beta\,\text{Dose}_{p(i)} + \gamma_1\text{Age}_i + \gamma_2\text{Male}_i + \gamma_3\text{Name}_i + \gamma_4\text{LogGDP}_{p(i)}\Big), \quad (4)$$

where $\Lambda(z) = e^z/(1+e^z)$ is the logistic c.d.f., $\text{Post2018}_i = \mathbf{1}[\text{year of naturalization} \geq 2018]$, $\text{Dose}_{p(i)}$ is the standardized enforcement dose of i's origin province, Age_i and Male_i are individual demographics, Name_i indicates adoption of a distinct Cambodian name, and $\text{LogGDP}_{p(i)}$ is the origin province's (standardized) log GDP. Standard errors are clustered by origin province and inference uses the score wild-cluster bootstrap. The coefficient β asks whether, among individuals who *did* naturalize, those from higher-enforcement provinces are disproportionately concentrated in the post-opening cohort, holding individual traits fixed. Because the sample is conditional on naturalizing, β describes the changing *composition* of buyers, not an individual's propensity to naturalize—the latter would require a sample of non-naturalizers (a population at risk), which returns one to the province-level count model of Section 5. The logit is thus a corroborating composition test, not an independent source of identification; the causal weight rests on the aggregate design.

Table 6 reports the estimates. A one–standard-deviation higher origin-province enforcement dose raises the odds of being a post-opening entrant by a factor of $\exp(0.838) = 2.31$ ($p < 0.01$; wild-cluster bootstrap $p < 0.001$). Origin income enters *negatively*, reproducing the aggregate finding that the margin is enforcement, not wealth. The post-opening entrants are younger and more male. A dose×male interaction is insignificant: the enforcement gradient does not differ by sex.

Table 6: Individual-level: origin enforcement and the post-2018 cohort (Logit)

DV: naturalized post-2018 (= 1)	(1)	(2) Heterogeneity
Origin enforcement dose (z)	0.838**	0.740***
	(0.266)	(0.221)
Dose × male		0.142
		(0.292)
Age at naturalization	−0.058***	−0.058***
Male	0.676***	0.692***
Adopted Cambodian name	−0.185	
Origin log GDP (z)	−0.446*	−0.495*
Estimator	Logit	Logit
SE clustered by province	Yes	Yes
Observations	2,195	2,195

Notes. Odds ratio on the enforcement dose in (1) is $\exp(0.838) = 2.31$. The dose coefficient has a score wild-cluster bootstrap p-value of < 0.001 (Table 7). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (a star denotes a 95% confidence interval excluding zero, or better).

6 Robustness and Inference

Wild-cluster bootstrap. Table 7 collects few-cluster inference. The score wild-cluster bootstrap on the PPML model returns $p < 0.001$ for the enforcement interaction, and the individual-level logit dose coefficient likewise returns $p < 0.001$. Both headline results survive the procedure recommended for our 28 clusters.

Alternative enforcement measures and the rank decomposition. Table 8 probes how the result depends on how the enforcement dose is built. Panel A shows it is stable across every *well-measured* version: total investigations per capita (baseline), the log count, and—directly testing the mechanism—the intensity of *low-rank* ("fly") prosecutions, defined as rank ≥ 7 or rank ≥ 9. All are positive and significant under the wild-cluster bootstrap. The lone exception is the *senior*-official ("tiger", rank ≤ 5) measure, which reverses sign. That measure, however, rests on only 479 senior officials nationwide and loads on small, peripheral provinces (Tibet, Ningxia, Qinghai) that send essentially no one to Cambodia, while Fujian—the principal sending province—recorded extensive broad enforcement but few elite purges. The reversal thus reflects the sparsity and small-province composition of senior-official prosecutions, not a failure of the broad-enforcement channel. Substantively, the response tracks the *breadth of local prosecution*—the dragnet reaching into local official–business networks—rather than high-profile elite takedowns. Panel B shows the effect survives excluding the qiaoxiang control and the top-three emigration provinces; dropping Fujian alone roughly halves the coefficient (to 0.496) but leaves it positive and significant, indicating that one province carries a substantial, though not exclusive, share of the identification.

Functional form and the margins. My estimand is a proportional effect on a count with many zeros, for which PPML is the appropriate, units-invariant estimator (Santos Silva and Tenreyro 2006; J. Chen and Roth 2024). A linear specification in arcsinh(count) yields a smaller, insignificant interaction, but J. Chen and Roth (2024) show why this comparison is uninformative: the average effect of treatment on arcsinh(Y) is not a percentage effect, and its magnitude depends on the arbitrary units of Y because it conflates the extensive and intensive margins. I therefore report the two margins separately. The *extensive* margin—a linear-probability model for whether a province records *any* naturalization in a year—is essentially zero (0.041, $p = 0.43$): enforcement does not govern *which* provinces participate, which is determined by pre-existing networks. Because this extensive-margin effect is null, conditioning on positive counts recovers the intensive margin without meaningful selection bias—the setting in which J. Chen and Roth (2024) note that the bounds on the intensive-margin parameter collapse to a point (their Remark 7 and footnote 31). That *intensive* margin—PPML among province-years with positive counts—is positive and significant (0.828, $p = 0.002$): among provinces with an established channel, higher enforcement scales the *volume* of exit. The effect thus operates entirely on the intensive margin, and the arcsinh null reflects that transformation's heavy weight on the (null) extensive margin rather than any fragility in the proportional effect.

Honest nulls. Two hypothesized heterogeneities do not appear in the data. Holders of a second investment passport (e.g. Vanuatu, St. Kitts; $n = 34$ among Chinese) are not especially concentrated in high-enforcement provinces, and the dose effect does not vary by sex. I report these as nulls.

Table 7: Few-cluster inference (score wild-cluster bootstrap, Rademacher, $B = 1999$)

Specification	Estimate	Cluster-robust p	Wild-cluster bootstrap p
Aggregate DiD (PPML), Eq. (1)	0.941	<0.001	<0.001
Individual logit (dose)	0.838	0.001	<0.001
Aggregate DiD, arcsinh outcome	0.192	0.23	0.20

Table 8: Robustness: alternative enforcement measures and samples

	Dose × Post-2018	Cluster SE	Wild-boot. p
Panel A. Alternative enforcement measures			
Total investigations, per capita (baseline)	0.941***	(0.254)	<0.001
Total investigations, log count	0.770***	(0.184)	<0.001
Low-rank "flies" (rank ≥ 7), per capita	0.888***	(0.234)	<0.001
Low-rank "flies" (rank ≥ 9), per capita	0.797***	(0.179)	<0.001
Senior "tigers" (rank ≤ 5), per capita	−0.833*	(0.361)	0.009
Panel B. Samples and controls			
Drop Fujian	0.496**	(0.190)	0.009
Drop top-3 emigration provinces	0.675**	(0.235)	0.004
Exclude qiaoxiang control	1.097**	(0.371)	0.003
Control for corruption index × post	0.880**	(0.293)	<0.001
Control for fraud-origin hub × post	1.054***	(0.256)	<0.001

Notes. Each row is the Dose × Post-2018 coefficient from a separate PPML estimation of Eq. (1), with province and year fixed effects, cluster-robust SE (province) in parentheses, and a score wild-cluster bootstrap p-value. Stars from the cluster-robust p: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. The senior-official ("tiger") measure rests on only 479 officials nationwide and loads on small, zero-emigration provinces; see text. The corruption-index control is the pre-campaign (2010–11) provincial MIMIC corruption index of H. Chen et al. (2018). The fraud-origin control is an indicator for whether a province contains a first-batch telecommunications-fraud "key rectification area" (*zhongdian diqu*)—a high-outflow fraud-origin area officially designated under the State Council Inter-Ministerial Joint Conference's "key area" rectification program (established October 2015) (Supreme People's Procuratorate of China 2015; Huang 2017). The indicator equals one for Hebei, Fujian, Jiangxi, Hunan, Guangdong, Guangxi, and Hainan—coded from the official designation, not from offender counts; judicial-record evidence independently confirms that such offending is spatially concentrated and Fujian-centered (Wu et al. 2025).

The estimates are consistent with a simple account: when domestic prosecution and expropriation risk rise, those exposed to it value a credible foreign exit option, and they exercise it when a route opens. The mechanism is given concrete form by documented cases in the same register: fugitives, money launderers, and US/UK-sanctioned operators who acquired Cambodian citizenship shortly before or after prosecution at home, several of whom also held second CBI passports (Kouprey 2026). Nine of ten Chinese nationals implicated in Singapore's 2023 billion-dollar money-laundering case held Cambodian citizenship.

The rank decomposition (Table 8) sharpens what kind of enforcement matters: the response tracks the *breadth* of local prosecution—overwhelmingly low-rank cases that reach into local official–business networks—rather than high-profile elite purges. This fits the documented profile of the buyers, who are connected businesspeople rather than senior officials themselves.

Deterrence versus the gray economy. Two readings of the broad-enforcement result are observationally close, and I cannot fully separate them. Under the first, intense local prosecution raises the risk perceived by the connected business class—whose patrons and associates are being investigated—and induces exit once a route opens. Under the second, low-level prosecution volume proxies the *size of a province's gray or patronage economy*, which independently generates both more corruption cases and more individuals with the means and motive to emigrate. I probe this directly by controlling for an *external* measure of the corruption stock: the provincial MIMIC corruption index of H. Chen et al. (2018), estimated independently for all 30 provinces from a wide set of indicators (the same authors provide a companion published estimate of the provincial shadow economy, (H. Chen et al. 2020)). Conditioning on the (pre-campaign) corruption index interacted with Post-2018 leaves the enforcement effect essentially unchanged—from 0.94 to 0.88, wild-cluster bootstrap $p < 0.001$ (Table 8)—even though the index is itself correlated with the enforcement dose ($r = 0.35$). The result is therefore *not* merely capturing provinces with larger corrupt economies. I read the estimates as consistent with a prosecution-driven exit channel, while noting two caveats: an alternative raw measure of procuratorial caseloads attenuates the effect more, and with so few high-volume provinces the design cannot fully separate the enforcement *shock* from the underlying corruption *level*.

Enforcement versus the scam economy. A related and sharper concern is that high-enforcement provinces are also the origin of operators in Cambodia's online-gambling and "scam-compound" economy, which boomed in Sihanoukville over 2017–2019; the enforcement gradient might then reflect a scam-network *pull* rather than a prosecution *push*. The worry is acute because Fujian is simultaneously a high-enforcement province and a documented telecom-fraud heartland. I address it with an external measure of fraud origin: the seven first-batch "key rectification areas" (*zhongdian diqu*) officially designated as high-*outflow* telecom-fraud origins under the State Council Inter-Ministerial Joint Conference's rectification program (established October 2015) (Supreme People's Procuratorate of China 2015; Huang 2017)—Fengning (Hebei), Longyan (Fujian), Yugan (Jiangxi), Shuangfeng (Hunan), Dianbai (Guangdong), Binyang (Guangxi), and Danzhou (Hainan). The coding thus follows the official designation, not any offender count. Independently, judicial-record studies of telecom-fraud offenders confirm the broader pattern that motivates the control—that such offending is spatially concentrated and overwhelmingly Fujian-centered (Wu et al. 2025).

7 Conclusion

Using the first individual-level register of investment naturalizations, I show that when Cambodia's 2018 reform opened an investment-citizenship route, take-up was disproportionately drawn from Chinese provinces that had experienced heavier anti-corruption enforcement—by a factor of roughly 2.6 per standard deviation of enforcement intensity, robust to wealth and network controls, to few-cluster inference, and reproduced at the individual level. Purchased foreign citizenship, in this setting, behaves like a hedge against domestic governance risk. More broadly, the result documents an unintended general-equilibrium margin of anti-corruption enforcement: pressure at home can displace wealth and people abroad, through a market in citizenship that is only beginning to be measured.

I emphasize what the design can and cannot show. It establishes that purchased citizenship is selected on home-province enforcement exposure, net of wealth and networks, and that this holds at both the regional and individual levels. It does not directly observe individual motive, and Cambodian naturalization is a thin, selected slice of total Chinese elite exit: the roughly 1,700 post-2018 Chinese naturalizations imply, even at the lower (pre-reform) investment threshold of about US$300,000, on the order of US$0.5 billion routed through this single, small channel—and several times that at the post-2018 threshold of nearly US$1 million—a lower bound that ignores residence-by-investment, family members, and fees, and a reminder that this is one narrow window onto a far larger outflow (Gunter 2004; Zucman 2013). The contribution is to make a mechanism the golden-passport literature has only theorized *visible*—at the level of the person and the home province.

References

Akcigit, U., S. Baslandze, and S. Stantcheva (2016). "Taxation and the International Mobility of Inventors". In: *American Economic Review* 106.10, pp. 2930–2981.

Alstadsæter, A., N. Johannesen, and G. Zucman (2018). "Who Owns the Wealth in Tax Havens? Macro Evidence and Implications for Global Inequality". In: *Journal of Public Economics* 162, pp. 89–100.

Beine, M., F. Docquier, and Ç. Özden (2011). "Diasporas". In: *Journal of Development Economics* 95.1, pp. 30–41.

Borjas, G. J. (1987). "Self-Selection and the Earnings of Immigrants". In: *American Economic Review* 77.4, pp. 531–553.

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2008). "Bootstrap-Based Improvements for Inference with Clustered Errors". In: *Review of Economics and Statistics* 90.3, pp. 414–427.

Chen, H., F. Schneider, and Q. Sun (2018). *Size, Determinants, and Consequences of Corruption in China's Provinces: The MIMIC Approach*. CESifo Working Paper 7175. CESifo.

— (2020). "Measuring the Size of the Shadow Economy in 30 Provinces of China over 1995–2016: The MIMIC Approach". In: *Pacific Economic Review* 25.3, pp. 427–453. DOI: 10.1111/1468-0106.12313.

Chen, J. and J. Roth (2024). "Logs with Zeros? Some Problems and Solutions". In: *Quarterly Journal of Economics* 139.2, pp. 891–936.

Cooley, A. and J. C. Sharman (2017). "Transnational Corruption and the Globalized Individual". In: *Review of International Political Economy* 24.5, pp. 732–753.

Griffin, J. M., C. Liu, and T. Shu (2022). "Is the Chinese Anti-Corruption Campaign Authentic? Evidence from Corporate Investigations". In: *Management Science* 68.10, pp. 7248–7273.

Gunter, F. R. (2004). "Capital Flight from China: 1984–2001". In: *China Economic Review* 15.1, pp. 63–85.

Huang, Zuhe (2017). "Causes and Prevention of Telecommunication Network Fraud". In: *Proceedings of the 2nd International Conference on Humanities Science and Society Development (ICHSSD 2017)*. Atlantis Press, pp. 164–173. ISBN: 978-94-6252-440-8. DOI: 10.2991/ichssd-17.2018.33. URL: https://doi.org/10.2991/ichssd-17.2018.33.

International Monetary Fund (2025). *Drivers and Effects of Residence and Citizenship by Investment*. IMF Working Paper WP/25/8. International Monetary Fund.

Kleven, H. et al. (2020). "Taxation and Migration: Evidence and Policy Implications". In: *Journal of Economic Perspectives* 34.2, pp. 119–142.

Kouprey (Mar. 2026). *Cambodian Citizenships Granted, 2000-2024*. Data compiled from Royal Gazette publications, 2000–late 2023. URL: https://mekongindependent.com/2026/03/cambodian-citizenships-granted-2000-2024/ (visited on 06/29/2026).

Le, Q. V. and P. J. Zak (2006). "Political Risk and Capital Flight". In: *Journal of International Money and Finance* 25.2, pp. 308–329.

Lin, C. et al. (2016). *Anti-Corruption Reforms and Shareholder Valuations: Event Study Evidence from China*. NBER Working Paper 22001. National Bureau of Economic Research.

MacKinnon, J. G. (2023). "Cluster-Robust Inference: A Guide". In: *Canadian Journal of Economics* 56.4, pp. 1239–1293.

Moretti, E. and D. J. Wilson (2017). “The Effect of State Taxes on the Geographical Location of Top Earners: Evidence from Star Scientists”. In: *American Economic Review* 107.7, pp. 1858–1903.

Munshi, K. (2003). “Networks in the Modern Economy: Mexican Migrants in the U.S. Labor Market”. In: *Quarterly Journal of Economics* 118.2, pp. 549–599.

National Bureau of Statistics of China (2012). *Tabulation on the 2010 Population Census of the People’s Republic of China*. Beijing: China Statistics Press.

— (2024). *China Statistical Yearbook*. Provincial gross regional product, various years; 2021–2023 revised per the 2023 Fifth National Economic Census. Beijing: China Statistics Press.

Organized Crime and Corruption Reporting Project (2024). *Criminals May Benefit as Cambodia Hides New Citizenship Data*. OCCRP. Accessed online.

Phnom Penh Post (2018). *Senate Approves Law on Citizenship*. Phnom Penh Post, 12 June 2018. New Law on Nationality replaced the 1996 statute and raised the investment requirement from approximately 1.25 billion riel (about US$312,000) to nearly US$1 million.

Pieke, F. N. et al. (2004). *Transnational Chinese: Fujianese Migrants in Europe*. Stanford University Press.

Rauch, J. E. and V. Trindade (2002). “Ethnic Chinese Networks in International Trade”. In: *Review of Economics and Statistics* 84.1, pp. 116–130.

Santos Silva, J. M. C. and S. Tenreyro (2006). “The Log of Gravity”. In: *Review of Economics and Statistics* 88.4, pp. 641–658.

Shachar, A. and R. Bauböck, eds. (2014). *Should Citizenship Be for Sale?* EUI Working Paper RSCAS 2014/01. European University Institute.

Sharman, J. C. (2017). *The Despot’s Guide to Wealth Management: On the International Campaign against Grand Corruption*. Cornell University Press.

Supreme People’s Procuratorate of China (2015). *Twenty-three Departments Jointly Combat New Types of Telecommunications Network Crime*. State Council Inter-Ministerial Joint Conference on Combating and Governing New Types of Telecommunications Network Crime, first meeting (October 2015). Establishes the inter-ministerial conference (23 departments, led by the Ministry of Public Security) and its “key area” (zhongdian diqu) rectification program. URL: https://www.spp.gov.cn/zdgz/201510/t20151011_105639.shtml.

Surak, K. (2021). “Millionaire Mobility and the Sale of Citizenship”. In: *Journal of Ethnic and Migration Studies* 47.1, pp. 166–189.

— (2023). *The Golden Passport: Global Mobility for Millionaires*. Harvard University Press.

Wang, Y. and B. J. Dickson (2020). *China Corruption Investigations Dataset*. Associated with “Coercive Capacity and the Durability of the Chinese Communist State”. DOI: 10.7910/DVN/9QZRAD.

Wu, Hequn, Ziwan Zheng, and Li Liu (Jan. 31, 2025). “The evolution and formation mechanism of the spatial distribution of cyber fraud offenders in China”. In: *Crime, Law and Social Change* 83.1, p. 14. ISSN: 1573-0751. DOI: 10.1007/s10611-025-10202-z. URL: https://doi.org/10.1007/s10611-025-10202-z.

Zucman, G. (2013). “The Missing Wealth of Nations: Are Europe and the U.S. Net Debtors or Net Creditors?” In: *Quarterly Journal of Economics* 128.3, pp. 1321–1364.

A Legal chronology and primary sources

This appendix documents the primary legal instruments governing Cambodian naturalization by investment, together with the operative provisions of the 2018 Law on Nationality. Dates and document numbers are taken from the original Khmer-language texts; the article summaries below are the author's translation from the Khmer and follow the original article numbering.

Table 9: Primary legal instruments: Cambodian investment naturalization

Instrument	Date	Effect
Law on Nationality (Royal Kram NS/RKM/0196/08)	24 Jan. 1996	Original Law on Nationality; permits naturalization by investment or donation.
Law on Nationality (Royal Kram NS/RKM/0618/008)	Nat. Assembly 31 May; Senate 11 June; promulgated 21 June 2018	Reformed the pathway: residence cut to 12 months (investment) / 6 months (donation); made an explicitly discretionary concession; monetary threshold delegated to sub-decree.
Sub-decree No. 112 ANKr.BK	30 Aug. 2018	Implementing regulation: set the operative investment amount (reported at nearly US$1 million) and procedures; operationalized the route.
Law on the Amendment of the Law on Nationality (Royal Kram NS/RKM/0925/030)	5 Sept. 2025	Further amendment (post-sample).
Sub-decree No. 226 ANKr.BK	2025	Implementing regulation repealing Sub-decree No. 112 (post-sample).

Operative provisions (2018 Law on Nationality, Ch. 6, "Naturalization by Investment"; author's translation).

- **Art. 18.** Investment naturalization is a discretionary concession of the state rather than a right of the applicant; the Prime Minister may waive conditions.
- **Art. 19.** Conditions include good conduct, no criminal-conviction record, residence, and competence in the Khmer language, script, history, and culture.
- **Art. 21 (investment route).** Requires legal residence of at least twelve months; the investment amount, in priority sectors, is set by sub-decree.
- **Art. 22 (donation route).** Requires legal residence of at least six months; the cash donation to the state budget is set by sub-decree.

Notes. (i) Monetary thresholds are fixed by sub-decree, not by the Law; the investment requirement was reported at nearly US$1 million following Sub-decree No. 112 (30 August 2018), up from roughly 1.25 billion riel (about US$312,000) under the prior regime. (ii) The route became operationally usable only when Sub-decree No. 112 took effect at the end of August 2018. (iii) The "4,000,000–10,000,000 riel" figure in Article 32 of the Law is a criminal fine for misuse of nationality documents—not an investment threshold. (iv) Publication of naturalization data in the Royal Gazette ceased after 2023.

B Province coding

Birthplace strings in the register are normalized to one of the 31 mainland provincial-level divisions via an explicit dictionary. Of 2,321 Chinese records, 94.7% map to a mainland province; 4.0% are Hong Kong/Macau SARs, 0.7% foreign birthplaces, and 0.6% are ambiguous (China but province not identifiable). The non-trivial mappings and the excluded categories are:

- **Misspellings → province:** Gaungdong → Guangdong; Tainjin → Tianjin; Jiansu → Jiangsu; Zhijiang, Zhujiang → Zhejiang; Hopei → Hebei; Shanndong → Shandong; Yuman → Yunnan; "Garg Su" → Gansu; "Shanxi/Shaanxi" → Shaanxi.
- **City/prefecture → parent province:** Fuzhou, Jinjiang, Anxi → Fujian; Guangzhou, Zhongshan, Nanshan → Guangdong; Liuzhou → Guangxi; Huzhou → Zhejiang; Funan → Anhui; Donghai → Jiangsu; Hokiang (Hejiang) → Sichuan.
- **Excluded as SAR:** Hong Kong, Macau.
- **Excluded as foreign:** non-China birthplaces (e.g. Mongolia, Vietnam, Thailand, Cambodia, United States, Canada) and Taiwan localities.
- **Excluded as ambiguous:** "China" (province unspecified) and blank entries.

Ambiguous and excluded records are dropped from the province panel. Because they are few and the estimates are robust to dropping high-leverage provinces (Table 8), the coding choices do not drive the results.